

Batch: 43 Roll No.: 16010324044
Experiment / assignment / tutorial No. 02 4
Grade: AA / AB / BB / BC / CC / CD / DD
Signature of the Staff In-charge with date

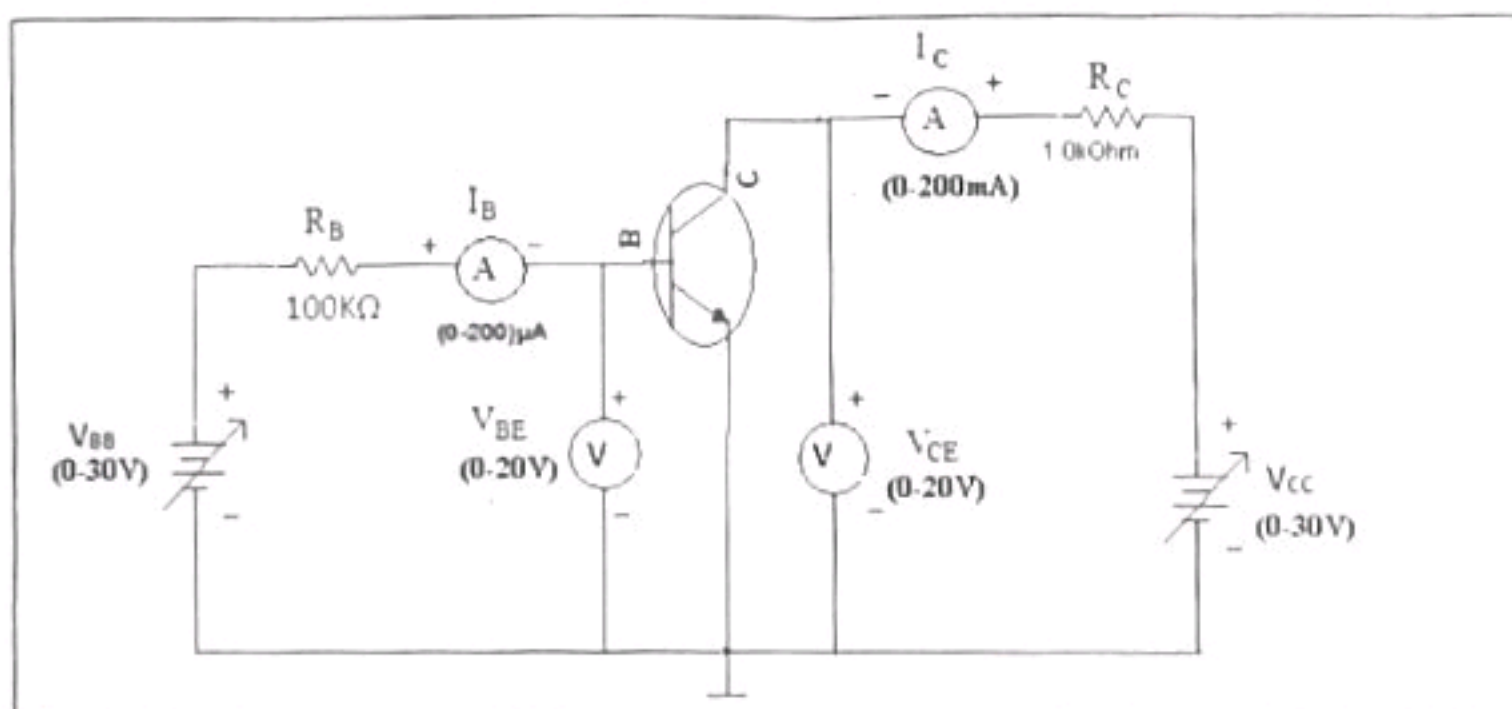
TITLE: To determine the input-output characteristics of BJT in Common Emitter configuration
AIM: Plot input and output characteristic for CE configuration using LTspice software

1. **OUTCOME:** Analyse the transistor circuits for DC operation

Procedure:

- 1) Build the setup for plot of input characteristics of BJT in common emitter mode in the workspace of LTspice. Select BC547B BJT from component tool, and simulate the circuit.
- 2) Perform DC sweep for different values of VCC using edit simulation command of software.
- 3) Make the setup for plot of input and output characteristics of BJT in common emitter mode.
- 4) Plot output characteristics (Collector Current (I_C) vs Collector emitter Voltage (V_{CE})) maintaining I_B Constant and measuring I_C and V_{CE} . I_B is maintained constant and I_C is varied using V_{CC} .
- 5) Plot input characteristics (Base Current (I_B) vs Base emitter Voltage (V_{BE})) maintaining V_{CE} Constant and measuring I_B and V_{BE} . V_{CE} is maintained constant and I_B is varied using V_{BB} .

Circuit Diagram: Setup (input/output characteristics)



Observation table 1:

Input characteristics plot:

P.T.O

Output characteristics plot:

P.T.O

Calculations:

V_{CE1}	V_{CE2}	I_{C1}	I_{C2}	$r_o = (V_{CE1} - V_{CE2}) / (I_{C1} - I_{C2})$	$r_o (Th)$
3V	6V	23.38 μA	53.38 μA	10K	

I_{B1}	I_{B2}	I_{C1}	I_{C2}	$\beta = (I_{C1} - I_{C2}) / (I_{B1} - I_{B2})$	$\beta (Th)$
22.87	5.55	7.50	1.29	358.54	

1. Explain the working of a transistor in CE configuration

In Common emitter (CE) configuration, The emitter terminal of the transistor is kept common for both input & output. The input is applied between the collector & emitter junction is forward biased, a small base current flows when a small input signal is applied. This small base current controls a much larger collector current because the collector-emitter junction is reverse biased.

Conclusion:

As a result, tiny change in base current ~~causes~~ a large change in collector current giving amplification.

Signature of Faculty with date

In CE configuration, the Transistor provides good current, voltage, and power gain. A small change in base current produces large change in collector current making it suitable for amplification and switching applications.

